stored in read-only memory (ROM) or EEPROM (electrically erasable programmable ROM) on the device. The same keys are then encrypted and packaged along with device identity information and stored in a central directory such as LDAP, where they can be accessed by API servers to authenticate and decrypt requests from clients or to encrypt responses to be sent to those devices. The architecture is shown in figure 12.9. A hardware security module (HSM) can be used to securely store the master encryption keys inside the factory to prevent compromise.

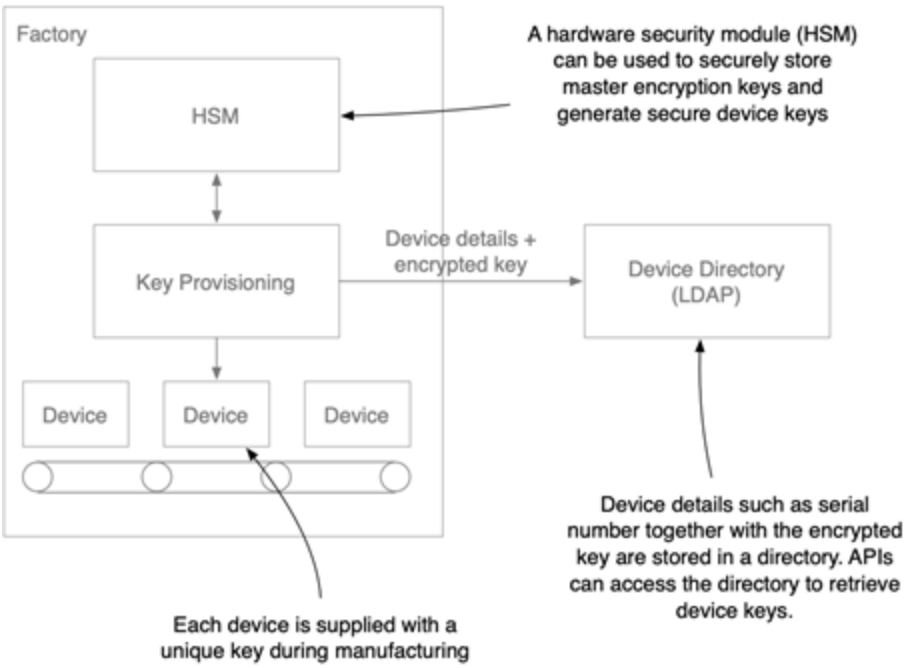


Figure 12.9 Unique device keys can be generated and installed on a device during manufacturing. The device keys are then encrypted and stored along with device details in an LDAP directory or database. APIs can later retrieve the encrypted device keys and decrypt them to secure communications with that device.